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PANDA-FES: Portable and Adaptable Neutron Diagnostics for Advancing Fusion Energy Science

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Abstract—Nuclear fusion is a potential source of carbon-free electricity with many concepts in development. The Portable and Adaptable Neutron Diagnostics for Advancing Fusion Energy Science (PANDA-FES) suite has been deployed since 2021 to measure neutron yield, energy, and spatiotemporal source location at two different Z-pinch fusion devices. This diagnostic can be used at a variety of facilities pursuing fusion in the magnetic, inertial, and magneto-inertial regimes. These different regimes have a wide range of time scales from less than 100 ns to a few μ s, neutron yields from 10^6 to 10^{11} , and noise environments. Neutron yield is measured through activation of ^{79}Br and ^{89}Y with calibrated detectors. Temporal, spatial and energy dependence of neutrons is measured with scintillators coupled to photomultiplier tubes. Experimental setups and data analysis methods have been developed for these conditions. Neutron yield, neutron energy anisotropy, and spatiotemporal evolution of the source have been measured.

Index Terms—neutron detectors, scintillation detectors, plasma pinch, fusion reactors

I. INTRODUCTION AND MOTIVATION

Nuclear fusion energy is a promising source of clean energy, which has recently been gaining increased interest from the world at large, as well as private industry [1], [2]. There are many different concepts to harness fusion energy that span from magnetic confinement fusion (MCF) to inertial confinement fusion (ICF), including an intermediate regime of magneto-inertial fusion (MIF). The large variation in fusion designs is accompanied by a large variation in the fusion timescales, yields and experimental conditions. Nonetheless,

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many of the fusion designs share the commonality that they plan to use D-T as the fusion fuel for power production. Given the difficulties of using radioactive tritium, many of the experiments use D-D to produce fusion in an experimental setting. When a deuterium plasma is sufficiently hot and dense, the ions fuse via the reactions, $\text{D} + \text{D} \rightarrow \text{T} + \text{H}$ and $\text{D} + \text{D} \rightarrow {}^3\text{He} + \text{n}$, which occur at nearly equal probability. As neutrons are a fusion product that is not significantly attenuated by metal, they can be detected outside a vacuum chamber in which fusion occurs. This feature makes them an ideal reference signal that can be consistently diagnosed over many fusion experiments. The energy and velocity of the fusion neutrons is determined from the kinematics of the reactant ions. Nominally the energy of D-D neutrons is 2.45 MeV, however this will shift depending on the temperature and velocity distributions of the ions. Understanding the neutron energy and angular distributions can inform whether a source is primarily thermonuclear fusion as opposed to beam-target fusion [3], which is vital to power scaling for energy production applications.

The Portable Adaptable Neutron Diagnostics for Advancing Fusion Energy Science (PANDA-FES) was designed at Lawrence Livermore National Laboratory (LLNL) and fielded at several fusion energy facilities starting in 2021. The LLNL team has collaborated with industry and university partners to gain understanding in the neutron production of fusion experimental devices being developed for energy production.

The PANDA-FES system was developed building on previous diagnostic work [3]–[5] to characterize the fusion plasma by measuring neutron yield, energy isotropy, and spatial and temporal distributions. The diagnostic suite consists of two detector types: lanthanum bromide (LaBr_3) activation detectors with yttrium caps and Eljen EJ-230 plastic scintillators coupled with photomultiplier tubes. The activation detector design was based on the National Ignition Facility Real-Time Neutron Activation Detectors (RTNADs) [6]–[8]. These activation detectors were fielded first at the MegaJoule Neutron Imaging for Radiography (MJOLNIR) dense plasma focus (DPF). Plastic scintillator detectors were fielded at the University of Washington (UW) [3]–[5], and several lessons learned were applied in the final design of PANDA-FES. Neutron sources with a range of yields from 10^6 to 10^{12} and time durations from tens of ns to a few μ s have been characterized with the PANDA-FES system.

Neutron yield measurements are necessary for calculating the fusion rate in a device. Potential issues with the yield

measurement may arise if there is neutron production at times not of interest during a device discharge, or if there is significant scattered neutron flux that may inflate the yield calculation. Yttrium has a higher threshold for the activation reaction that is used to reduce sensitivity to scattered neutron signal. Neutron detectors with sensitivity to event timing are fielded in conjunction with activation detectors to provide an understanding of the emission duration.

Scintillator-based detectors can be operated in either continuous mode or pulse counting mode and placed in flexible spatial configurations close or far from a device. Neutron time-of-flight (nTOF) measurements and individual event measurements can be used depending on the neutron flux which is determined by neutron yield, detector distance, and neutron emission duration. For a high yield and short emission time, the neutrons (and x-rays) will arrive at the detector over a time on the order of the instrument response time, meaning a continuous signal is observed proportional to the radiation emission. The neutron energy can be calculated using time-of-flight when the scintillators are operated in continuous mode or through accumulating a histogram of pulse integrals when they are operated in pulse counting mode. When the neutrons are emitted over a longer time and the flux is low because of larger detector distance or lower yield, single event pulses can be measured.

The distribution of neutron energies is important for understanding what type of fusion reaction is happening inside the device. Thermonuclear fusion is a spatially isotropic reaction that outputs 2.45 MeV neutrons for D-D fusion. Beam-target fusion occurs when an ion beam is accelerated into a relatively dense and stationary target material. D-D beam-target neutrons are forward directed and characterized by an up-shifted neutron energy in the forward direction, and down-shift in the rear direction [9]. When the flux on the scintillator is lower, for example on a lower yield device or when the detectors are placed farther away or the radiation emission is spread over a longer time, the detectors can be operated in pulse counting mode.

The PANDA-FES diagnostic system was fielded on staged Z-pinch experiments at the CESZAR [10] linear transformer driver at the University of California San Diego (UCSD) and the FuZE [11] sheared flow stabilized (SFS) Z-pinch device at Zap Energy. The CESZAR device has a tens of ns long radiation emission time with a neutron yield on the order of 10^7 . The FuZE device has a radiation emission time on the order of $2 \mu\text{s}$ with neutron yields up to 10^8 neutrons into 4π . The spatiotemporal source location measurement, neutron energy anisotropy measurement, and nTOF measurements were developed at these facilities with improved experimental setups, data analysis codes, and Monte Carlo modeling capabilities. The full range of operation in terms of neutron yield and radiation emission time scales are shown in Figure 1 for a detector distance of 20 cm from a point source. The results shown in this paper from the fusion energy platforms FuZE and CESZAR are in the middle of the range of capabilities.

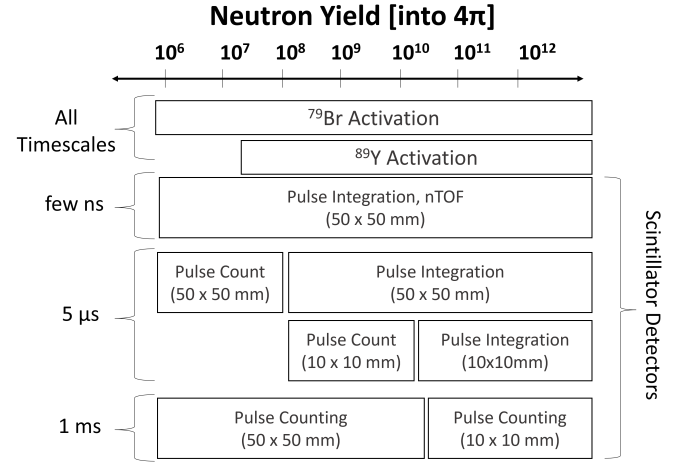


Fig. 1. The PANDA system can be fielded over a wide range of neutron yields and timescales by using different modes of operation and different sizes for the scintillator detectors. The Neutron yield here assumes a detector distance of 20 cm from a point source.

II. NEUTRON YIELD MEASUREMENTS WITH ACTIVATION DETECTORS

A. Methodology and Experimental Hardware

The commercially available LaBr_3 gamma spectrometer detectors were purchased from AMETEK [12]. A 1.5-inch diameter by 1.5-inch length cylinder of $\text{LaBr}_3(\text{Ce})$ crystal is coupled to a 2-inch diameter photomultiplier tube (PMT) and an Ortec Digibase-E module. The Digibase-E records list mode data [13], meaning that both time and energy are recorded. A 1-cm thick yttrium cap is attached around the outside of the LaBr_3 scintillator, shown in Figure 2. Neutrons inelastically scatter on ^{79}Br and ^{89}Y isotopes and the de-excitation of the metastable isomers is recorded over 10 half-lives to calculate the neutron fluence. The decay energy and half-life for $^{79\text{m}}\text{Br}$ is 208 keV and 4.8 s, and for $^{89\text{m}}\text{Y}$ they are 909 keV and 15.7 s, respectively. The cross sections of these reactions is shown in Figure 3. There is some disagreement between the different evaluations, but these cross sections are not directly used in this analysis for D-D neutron sources with neutron energy approximately the same as the calibration source.

The detector is placed inside of an aluminum tube with feedthroughs in the back for signal/power (ethernet) and trigger (BNC) cables. Because the detector and cables are shielded from electromagnetic interference (EMI), as shown in Figure 4, they can be placed close to pulsed power devices. The list mode data are transferred to a Digital Acquisition (DAQ) laptop and saved using the Maestro software [14]. The detector is powered to a bias voltage of nominally 750 V using a power over ethernet network switch. A 150 second transistor-transistor logic (TTL) pulse from a delay generator gates the detector count time.

B. Data Analysis for Activation Detectors

Raw data are recorded by the MAESTRO multichannel analyzer (MCA) emulation software in list mode with the energy channel and time of each event, shown in Figure 5. These

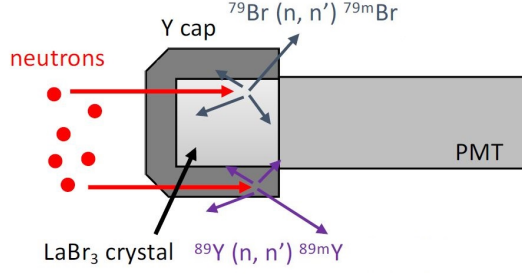


Fig. 2. Cartoon of the activation detector. The LaBr scintillator detects the decay from ^{79m}Br and ^{89m}Y which are excited by neutrons passing through the scintillator itself and the yttrium cap, respectively. The decay energy and half-life for ^{79m}Br is 208 keV and 4.8 s, and for ^{89m}Y they are 909 keV and 15.7 s, respectively.

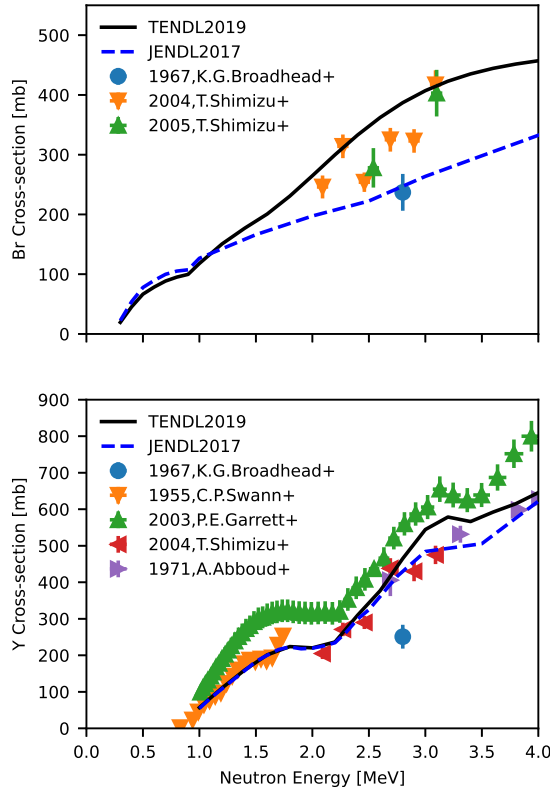


Fig. 3. Calculated [15], [16] and experimentally measured [17]–[22] cross-sections for (top) $^{79}\text{Br}(n,n')^{79m}\text{Br}$ and (bottom) $^{89}\text{Y}(n,n')^{89m}\text{Y}$ reactions.

list mode data are processed using custom LLNL-developed Python software. A range of channels that correspond to the ^{79m}Br and ^{89m}Y decay lines are integrated and the decay over time is fit using an exponential function, shown in Figure 6. The time dependent background determined by the adjacent energy channels is simultaneously fit to an exponential plus a constant function so it can be subtracted from each spectrum.

The background is fit with a single exponential decay with magnitude A_{bg} and decay constant τ_{bg} , and a constant background amplitude of C_{bg} . From this, the number of

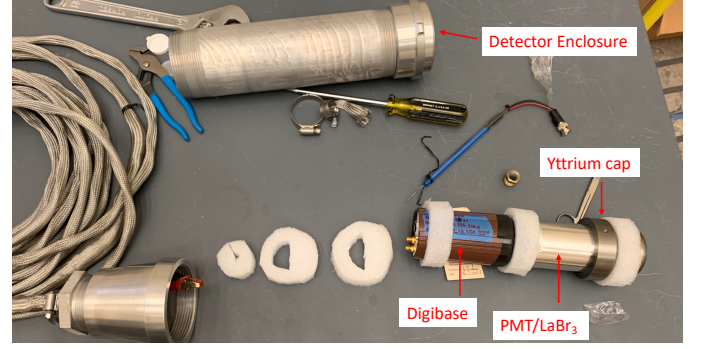


Fig. 4. Photograph of the LaBr activation detector and enclosure.

background counts per bin is given by $F_{bg}(t)$, where t is the start of the bin and Δt is the bin width:

$$F_{bg}(t; C_{bg}, A_{bg}, \tau_{bg}) = C_{bg}\Delta t + A_{bg}\tau_{bg} \left[\exp\left(-\frac{t}{\tau_{bg}}\right) - \exp\left(-\frac{t + \Delta t}{\tau_{bg}}\right) \right]. \quad (1)$$

The signal is fit with a exponential decay based on the known half-lives, $\lambda_{1/2} = \tau \ln 2$, where τ is the decay constant of an isotope. The expected counts per bin is given by F_{sig} .

$$F_{sig}(t; A_0) = A_0\tau \left[\exp\left(-\frac{t}{\tau}\right) - \exp\left(-\frac{t + \Delta t}{\tau}\right) \right]. \quad (2)$$

The total activation, C_0 , is given by:

$$C_0 = \sum F_{sig}(t) = A_0\tau. \quad (3)$$

The fits to the background and signal are optimized together using a χ^2 minimization of

$$\chi^2 = \sum \frac{(Y_{bg} - F_{bg})^2}{\sigma_{bg}^2} + \sum \frac{(Y - F_{bg} - F_{sig})^2}{\sigma_{sig}^2}, \quad (4)$$

where n is the number of time bins, Y_{bg} is the signal within the background region, Y is the signal within the signal region, and σ represents the uncertainty in the fits. To perform this fit we use the *scipy.optimize.curve_fit* function in python [23]. This method uses the generic minimization of χ^2 , and thus requires a constant for σ_{bg} and σ_{sig} . First these constants are defined as $\sigma_{bg}^2 = Y_{bg}$ and $\sigma_{sig}^2 = Y$, then the result of this fit is used as the updated σ for a second fit. The *curve_fit* function calculates a covariance matrix that is then used to find the uncertainty in A_0 , σ_{A_0} . σ_{A_0} can be used to find the uncertainty in C_0 : $\sigma_{C_0} = \sigma_{A_0}(C_0/A_0)$.

A calibration constant, discussed in the following section, is used to convert from detector counts to neutron fluence at the detector position. Then, the distance from the source is used to calculate the neutron yield of the device, assuming a point or line source geometry and isotropic neutron emission into 4π . The analysis code displays the raw list mode data, the decay calculation, and some background lines to verify the operation of the detector.

With an absolute calibration, neutron yield is measured from D-D fusion. Neutron yields have been measured on the FuZE device and the CESZAR device, where parameter scans of device conditions are used to optimize neutron yield. For

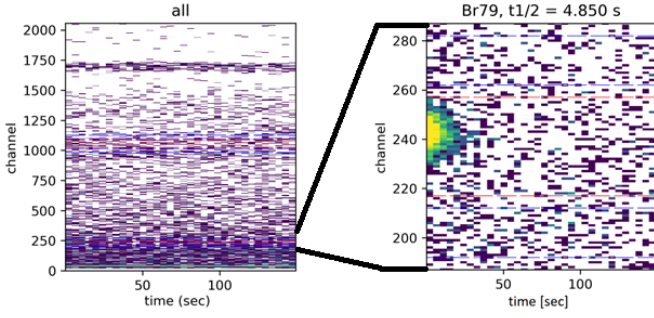


Fig. 5. Total list mode raw data (left) and a zoom in on the channels that correspond to the Br79 activation line and background (right).

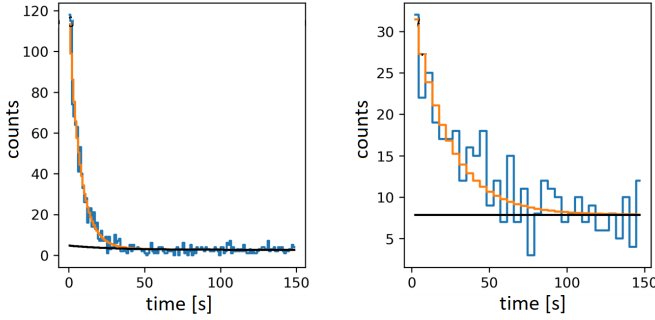


Fig. 6. Exponential decay fit in orange for the activated isotopes Bromine-79 (left) and Yttrium 89 (right) used to calculate neutron yield. Fit to the background is in black.

example, with the detector 61 cm from the plasma center a fluence of $8.47 \times 10^2 \pm 1.19 \times 10^2$ n/cm² was measured from CESZAR where we calculated a yield of $3.96 \times 10^7 \pm 6.34 \times 10^6$ (16%) neutrons into 4 π from a point source; and with the detector 27.7 cm from the plasma center on FuZE a flux of $5.92 \times 10^3 \pm 2.30 \times 10^2$ n/cm² was measured where we calculated a yield of $5.38 \times 10^7 \pm 6.45 \times 10^6$ (12%) neutrons into 4 π from a 32 cm line source. Additional work is planned using the PANDA-FES activation detectors at FuZE, CESZAR, and other prospective fusion energy concepts.

C. Diagnostic Calibration

The LaBr activation detectors were cross-calibrated with a beryllium activation detector, using the deuterium filled MJOLNIR [24] DPF at LLNL as a neutron source, with the setup shown in Figure 7. Three LaBr activation detectors were co-located with a beryllium detector that was calibrated at Sandia National Laboratories Ion Beam Laboratory (IBL) [25]. The estimated uncertainty in the calibration of the beryllium detector is 3%. Data were recorded on all activation detectors over a range of neutron yields. A calibration constant, shown in Figure 8, was calculated for both ⁷⁹Br and ⁸⁹Y lines for each detector and applied in post-processing to convert the number of counts to neutron fluence. The extracted calibration constants are shown in Table I and II. The uncertainty in the position of the detectors and the fit of the decay are propagated into the uncertainty in the calibration constant. The total uncertainty on the calibration constant is 6%, including

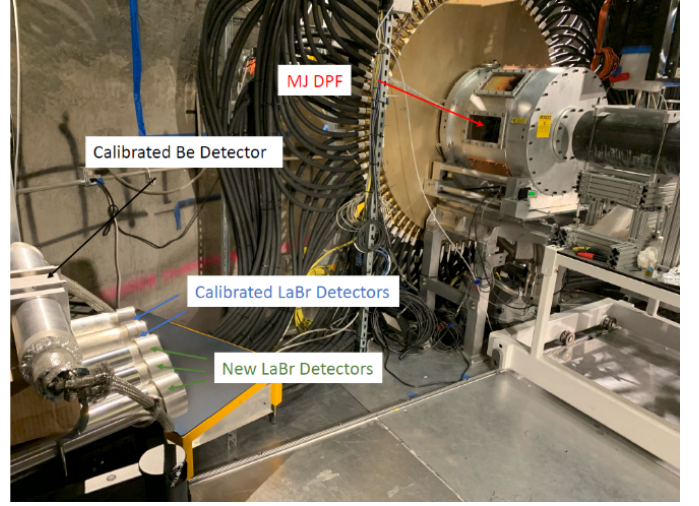


Fig. 7. The MJOLNIR DPF is used as a neutron source for the cross calibration of LaBr activation detectors with a calibrated Beryllium activation detector.

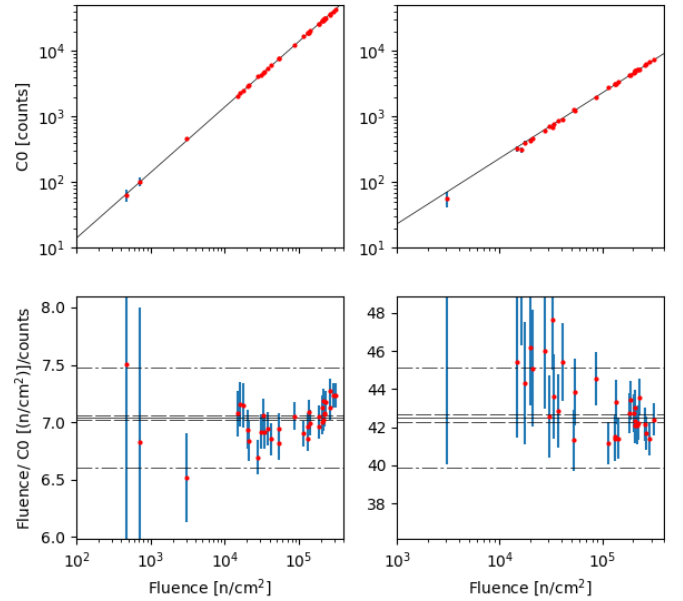


Fig. 8. The C0 fit parameter for ⁷⁹Br (left column) and ⁸⁹Y (right column) is linearly related to the incident neutron fluence calculated using the Be detector yield. The proportionality constant (bottom row) between fluence and C0 is the calibration constant for one of the three LaBr detectors that were cross-calibrated.

the systematic uncertainty from the Beryllium calibration and location as well as statistical uncertainty from the fit of the decay in the ⁷⁹Br and ⁸⁹Y lines. The lower level sensitivity of these activation detectors is 5×10^2 n/cm². The maximum count rate is 50k counts per second and is limited by the digiBase-E onboard memory size and the data transfer frequency between the digiBase-E and the data acquisition (DAQ) computer.

The neutron yield measurement can be affected by the neutron energy, because the neutron cross sections are energy dependent. The probability of neutrons activating ⁷⁹Br and ⁸⁹Y increases as neutron energy increases. The validity of

^{79}Br	$C = \phi/C_0$	σ_C/C	σ_{fit}/C	$2\sigma_{D_i}/D_i$	$2\sigma_{D_{Be}}/D_{Be}$	$\sigma_{Cal_{Be}}/Cal_{Be}$
AD1	7.04	6.1%	0.30%	3.92%	3.70%	3.00%
AD2	7.06	6.1%	0.30%	3.87%	3.70%	3.00%
AD3	7.14	6.1%	0.31%	3.82%	3.70%	3.00%

TABLE I

 ^{79}Br CALIBRATION FACTORS AND UNCERTAINTIES FOR THE THREE LABR DETECTORS.

^{89}Y	$C = \phi/C_0$	σ_C/C	σ_{fit}/C	$2\sigma_{D_i}/D_i$	$2\sigma_{D_{Be}}/D_{Be}$	$\sigma_{Cal_{Be}}/Cal_{Be}$
AD1	42.49	6.1%	0.50%	3.92%	3.70%	3.00%
AD2	42.30	6.1%	0.50%	3.87%	3.70%	3.00%
AD3	43.79	6.1%	0.52%	3.82%	3.70%	3.00%

TABLE II

 ^{89}Y CALIBRATION FACTORS AND UNCERTAINTIES FOR THE THREE LABR DETECTORS.

the calibration constants extracted in this work rely upon an assumption that the neutron energy of the unknown source is comparable to that of the source used during calibration.

The neutron energy of the MJOLNIR DPF used for cross calibration is expected to be anisotropic and varies with neutron yield. The Be activation detector and LaBr_3 activation detectors have different energy dependent efficiencies, so if the energy of the neutrons emitted from the DPF changes, the measured count rate in the two detectors will be affected differently. Because the slope of the ^9Be cross section is steeper than that of ^{79}Br , the efficiency of the Be activation detector will increase more for a given change in neutron energy. The calibration constant is valid for D-D fusion neutrons of energy around 2.45 MeV, and a new calibration would need to be performed to apply this diagnostic to systems with a different neutron energy. The change in the cross section, which directly influences the detector efficiency, is 4% for ^{79}Br for +/- 100 keV change in neutron energy about 2.45 MeV.

The DPF neutron emission is also angle dependent, with a higher neutron fluence in the forward direction of the device. Based on simulations, this effect could be on the order of 10% over a 10 degree change in angle [26]. The effect of the DPF's anisotropy in both neutron energy and neutron fluence was minimized by placing the two detectors at the same polar angle.

III. ENERGY AND TIME RESOLVED NEUTRON MEASUREMENTS WITH PLASTIC SCINTILLATORS

A. Experimental Hardware and Setup

EJ-230 plastic scintillators [27] were coupled to 2-inch-diameter Hamamatsu H1949-51 assembly with R1828-01 photomultiplier tubes (PMT) and voltage dividers [28]. The scintillator plastic and PMT were chosen to optimize the time response, dynamic range of the measurement, and avoid PMT saturation. The detectors were assembled by Eljen Technology so that the scintillators can be easily dismounted from the PMTs; this allows different sizes of scintillators to be used and neutral density (ND) filters to be inserted to reduce the light signal on the PMTs to avoid saturation. Scintillator sizes of 5×5 cm and 1×1 cm (diameter $\times$ depth) were used; this allowed for a variation of 125 times in volume to be chosen depending on the experimental setup. ND filters of 10 to 100 fold light attenuation were used. A mu metal shielding reduces sensitivity to external magnetic fields. Detectors require more

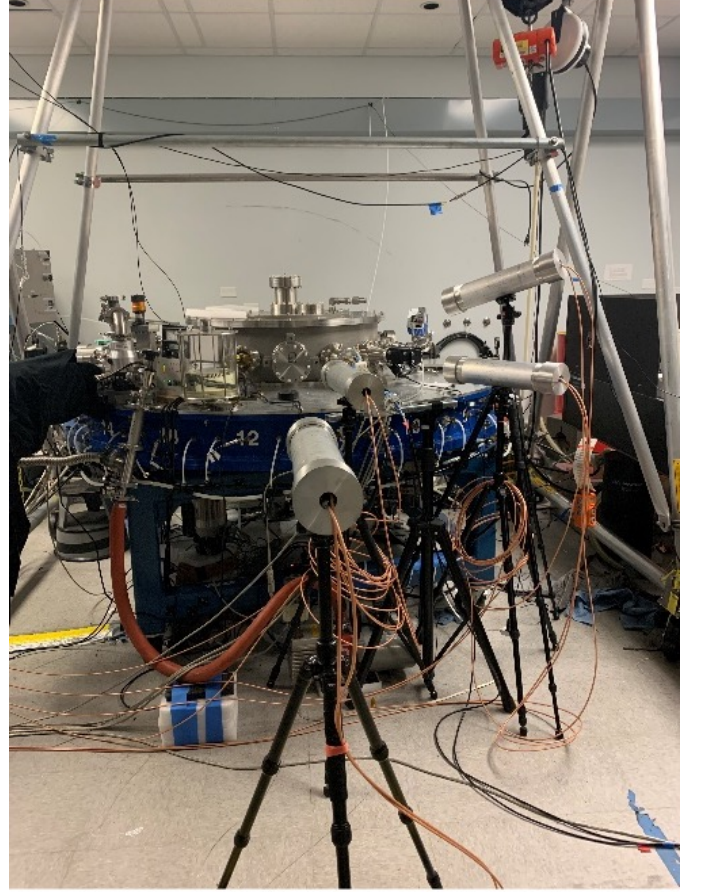


Fig. 9. Three plastic scintillator detectors were placed at 1 m from the center of the CESZAR device at different angles, and one plastic scintillator detector was placed at 3 m flight path to measure anisotropy and neutron energy.

EMI shielding at facilities that have high noise levels, so the plastic scintillator detectors were placed inside aluminum tubes, as shown in Figure 9. Lead was used for photon attenuation at facilities with high x-ray flux. The effects of neutron and photon attenuation in the shielding elements can be modeled using the Monte Carlo N-Particle radiation transport code (MCNP6) [29].

The detector setup is tailored to either pulse counting or continuous mode data acquisition by controlling the flux incident on the scintillator. An oscilloscope is used to record the detector signal for a preset amount of time during and following the device discharge. This time can be set on the

order of 100s of ns to 100s of μ s. This digitized time trace is saved and post-processed using custom data analysis codes. The Tektronix oscilloscope MSO58LP with 1 GHz bandwidth limits the signal level from the PMTs that can be digitized, with a maximum ± 10 V for a $50\ \Omega$ input [30]. The detector setup was chosen to keep the signal level within a range measurable by the oscilloscope and below the saturation limits of the PMT, as discussed in the following PMT linearity measurements section. A 1 cm diameter by 1 cm height right cylinder or a 5 cm diameter by 5 cm height right cylinder organic plastic scintillator can be coupled with the PMT to change the count rate. For the two scintillator sizes used so far, the efficiency is proportional to the scintillator volume. The count rate is also inversely proportional to the distance from the source squared. To operate in pulse counting mode, the count rate must be low enough to distinguish between individual radiation events. In continuous mode, when pulses are piled up to such a degree that a smooth continuous signal is measured, the signal level can be adjusted by changing the PMT gain and by adding a ND filter between the scintillator and PMT to reduce light onto the photocathode. By using these different levers, the PANDA-FES scintillator detectors have been fielded at a variety of devices with different yields and emission time scales.

The spatial setup of the PMTs has been used to measure anisotropy or the spatial extent of a source. Detectors are placed at different angles relative to the Z-pinch axis to measure anisotropy either in pulse counting mode or continuous mode. The plasma emission length has been measured on a sheared-flow stabilized z-pinch device by placing detectors along the emission region close to the device chamber.

B. Data Analysis with Plastic Scintillator Detectors

Data from two detectors at different distances in the same discharge from the FuZE device are shown in Figure 10 to illustrate pulse counting and continuous modes of operation. In pulse counting mode, peak finding is used to locate and isolate individual events as shown in Figure 10a. The integral of each pulse, which is proportional to the light generated by the scintillator, can be calculated by fitting the pulse with a function in which the pulse shape is parameterized by the amplitude, rise time, and fall time of the signal. The goodness of fit for each event is calculated using a reduced chi-squared metric, as described in detail in [3]. When the scintillation pulses are close together in time, the integrals can be extracted by fitting two overlapping pulse functions. However, there are cases where two events are detected too close in time to separate. The limit of the pileup recovery in this analysis is 20 ns. Pulses with a bad fit and poor chi-squared metric are rejected.

In continuous mode, shown in Figure 10b, the detector signal as a function of time is recorded. Unless the PMT is saturated, the signal level is proportional to the energy deposited by incident radiation. In a high flux experiment, the PMT can become saturated either by having too high of a voltage that causes space-charge effects or by depleting the charge in the capacitors within the PMT [31]. PMT saturation

is avoided in these experiments by limiting the amount of light incident on the photocathode. The saturation limits of these PMTs were measured and are discussed in the following section.

Assuming that one event produces both photons and neutrons simultaneously and the radiation burst is over a short time (few 10s of ns), then the time of flight of neutrons can be used to determine the neutron energy. If multiple peaks are observed, they can be characterized as x-rays or neutrons using detectors at different distances from the source. X-rays travel at the speed of light and arrive at a given detector before the neutrons when detectors are placed at a large enough distance from the source. The time difference of x-ray arrival at a shorter vs. farther distance is much smaller compared with that of neutrons. The arrival time of neutrons at detectors at different distances is used to calculate neutron velocity, which is related to kinetic energy. This time-of-flight analysis can be complicated by more than two peaks or peaks overlapping in time.

There can be significant scattered neutron components to the measured signal, which can be subtracted using a shadow bar setup in which a pair of detectors are placed as close together as possible while blocking direct signal to one of the detectors with a thick piece of high density polyethylene so that it measures the isolated scattered signal and the other measures the combined direct and scattered signals. The different detectors in this diagnostic are cross-timed and gain matched so that the signals can be compared among the set of identical detectors.

The detector is configured (by varying the distance from source, bias voltage, ND filtering, and/or scintillator size) such that the radiation flux produces a smooth continuous signal from the scintillator. The cross-timed and gain matched detectors can be placed at different axial or azimuthal locations to measure the spatiotemporal radiation signal of the neutron producing plasma. These different modes of data analysis have been developed to achieve multiple experimental objectives of estimating deuteron beam energy, calculating neutron energy from neutron TOF, and spatiotemporal source distribution measurement.

C. Plastic Scintillator Diagnostic characterization

1) *PMT gain matching:* Given the strong dependence of PMT gain on bias voltage, even slight differences in manufacturing can result in different gain in different PMTs with the same supply voltage. Thus, in order to aggregate data across multiple detectors, we quantify the relative gain of all the detectors. This was done using a known gamma source of either ^{137}Cs or ^{60}Co to record data in pulse counting mode and create a histogram of pulse integral spectra. These data are used to adjust the bias voltage prior to the experiment so that all detectors operate at similar gain, and is used to calculate correction factors that are used when post-processing the data. This allows data from multiple detectors to be aggregated and the measurement time is greatly reduced compared to a single detector [5].

The scintillator geometry is modeled in MCNP6 to simulate the energy deposited in the scintillator material from the

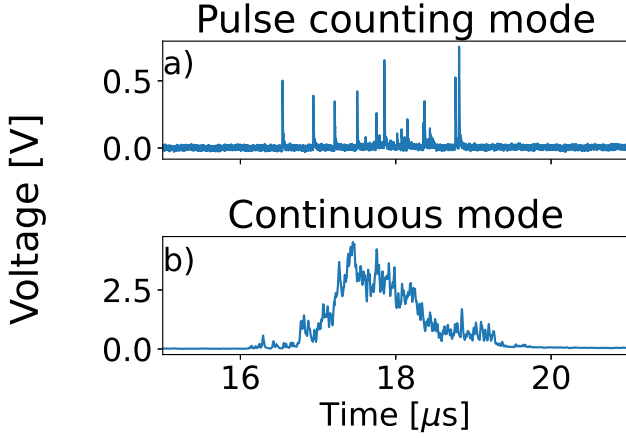


Fig. 10. Example of pulse counting mode (a) and continuous mode (b) for plastic scintillator detectors placed far and close to the FuZE device, respectively.

gamma source used during gain matching. This spectrum is broadened to match the experimental data, and the Compton edge of the simulated Compton spectrum is found. An example of the fitting procedure is shown in Figure 11, where the Compton edge location, amplitude, and the detector energy resolution broadening are fit parameters. The lower part of the spectrum is excluded from the fit because the small integral noise pulses from the PMT are not modeled in the simulation of the scintillator. Any differences in PMT gain will be observed as shifts increasing or decreasing the pulse integral, or charge, reflected in the location of the Compton edge. The Compton edge location fit parameter provides a quantifiable metric for the detector gain, used to track the gain shift in Figure 12. A gain correction factor is calculated for small differences in PMT gain and applied as a multiplier to the signal in continuous mode or the pulse height threshold in pulse counting mode. In this way, the detectors are gain matched as close as practical and corrected for small differences in PMT gain to increase the accuracy of this method.

Gain drift was studied by looking at the gain at different times from the initial gain matching. The relative gains are calculated using Compton spectra from known gamma sources. As shown in Figure 12, the Compton edge location was observed at 29 and 176 days after the initial gain matching. We find drifts on the order of a few percent, with peak values up to 8%.

2) *PMT linearity measurements:* Previous experiments have shown PMT saturation with μs -long radiation discharges. The PMT runs out of charge when the internal capacitors are depleted by high signal with insufficient recovery time. There is an intermediate region where the signal can be unpredictable followed by a complete shutoff of signal from the PMT if the incident radiation continues [31]. The PMTs purchased for the new diagnostic system were chosen with a larger capacitance and a faster internal circuit to limit saturation. Tests were done to quantify the signal level where the PMT starts to saturate.

The Hamamatsu R1828 PMT [28] was placed in a dark

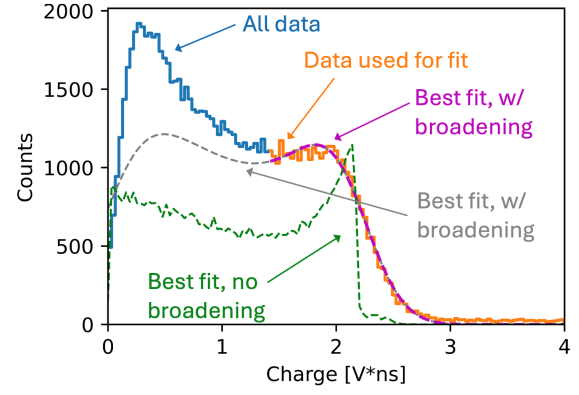


Fig. 11. A simulated pulse integral spectrum was fit to measured Compton spectra from a ^{137}Cs source. The fit includes Gaussian broadening to best match the data, excluding lower energies. The portion of the spectrum used in the fit has good agreement between the simulation (purple) and experimental data (orange). The lower energy part of the spectrum for the simulation (gray) and experimental data (blue) are not part of the fit.

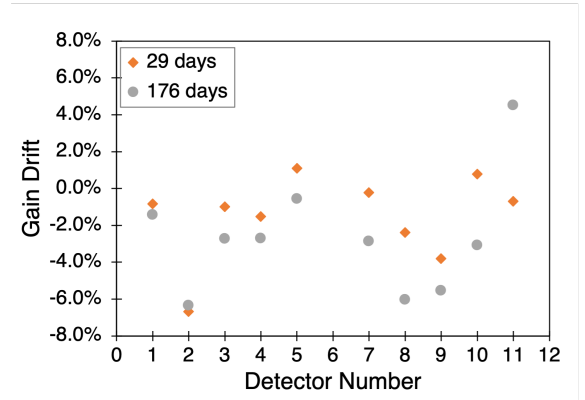


Fig. 12. Gain drift quantified by the location of the measured Compton edge for multiple detectors taken at 29 and 176 days after the initial gain matching. The values of the mean are -1.51% and -2.66% for 29 and 176 days, respectively. The standard deviation is 2.30% and 3.18% for 29 and 176 days, respectively.

box with a light-emitting diode (LED) and an avalanche photo diode (APD). A bias voltage of -1800 V was applied to the PMT for these tests. The APD is less sensitive to saturation and was used to measure the light output from the LED. The light incident onto the photocathode of the PMT was varied by changing the voltage applied to the LED and the amount of neutral density (ND) filters on the face of the PMT. The LED was turned on for 5 μs and the voltage output by the PMT and APD were measured on an oscilloscope. The light level was increased until 1.86 V was output from the PMT with no observed saturation on the 5 μs time scale, which is the maximum time scale for the planned experiments with this diagnostic. The PMT is in linear mode up to 9.3 V $\cdot\mu\text{s}$, equivalent to 186 nC. This is shown in the linear relationship between the signal from the PMT (with corrections applied for ND filters) and the APD, as shown in Figure 13 during expected time and signal levels. To observe saturation on this PMT, the time scale of the test was increased until the PMT signal deviated from the normalized APD signal shape as shown in Figure 14. The measurement with a longer exposure

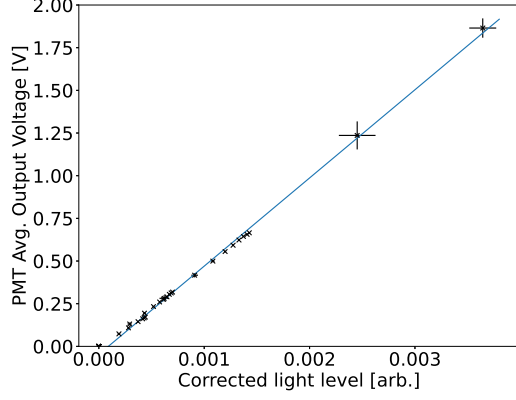


Fig. 13. The linear relationship between the PMT signal and the corrected “actual” light level measured using the APD shows that the PMT is in linear mode for the range of signals that were tested, up to $9.3 \text{ V} \cdot \mu\text{s}$, equivalent to 186 nC .

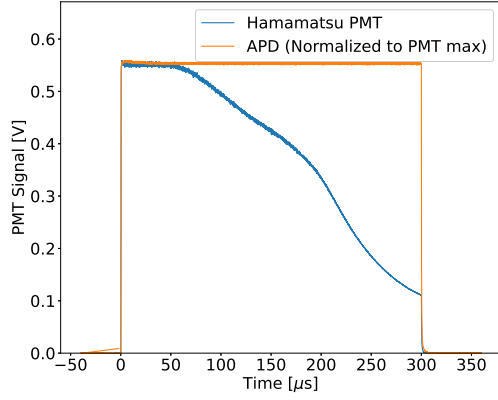


Fig. 14. The signal from the PMT drops off compared with the actual light signal measured by the APD when the illumination time is increased, showing that the PMT has run out of charge.

time predicts a charge saturation limit of $27.5 \text{ V} \cdot \mu\text{s}$, equivalent to 550 nC , from maintaining linear behavior at 550 mV for $50 \mu\text{s}$ into a 50Ω terminated oscilloscope.

D. Selected Results from Scintillator Detectors at FuZE

The μs -long time duration of radiation emission from the FuZE device made pulse counting mode ideal for measuring the neutron spectrum from the fusion plasma. The distance of the detectors was chosen to balance maximizing the number of neutron events per discharge while mitigating against pulse pileup. The integral value of all pulses were aggregated from the multiple detectors at the two angles and separately histogrammed to calculate the neutron energy spectrum at each angle.

This new diagnostic improves on previous experiments [3], [5] with an increased number of detectors and detector types. The PANDA-FES systems deployed at UCSD and Zap Energy have eight and sixteen scintillator detectors, respectively, enabling spatial measurements for each discharge of a z-

pinch device. The scintillator is faster than the previously used EJ-204 [32] and the PMT voltage divider holds more charge than the previously used Hamamatsu R7724 PMTs [28]. The gain matching method allows for direct comparison of the scintillator detector signals without swapping detector locations and repeating the measurement. For measurements in pulse counting mode, detectors placed in an array at one location are added together, improving counting statistics. With a large number of detectors, different measurements of the anisotropy and spatial source can be done in parallel. Compared with previous work, the deuteron beam energy measurement detailed in [3] was accomplished in 46 shots instead of 116 shots, reducing the time needed to obtain a result by a factor of 2.5. This factor of 2.5 is lower than would be expected given the number of detectors employed, and is due to the faster (order of $2 \mu\text{s}$) neutron emission at Zap Energy, compared to around $5 \mu\text{s}$ at UW.

E. Selected Results from Scintillator Detectors at CESZAR

The short time duration of the radiation emission from the CESZAR device made it an ideal candidate for nTOF measurements with the plastic scintillator detectors in continuous mode. The setup of this system is shown in Figure 9. There is a hard x-ray signal seen on most shots that was reduced using lead shielding so that the signal level was the same order of magnitude as the neutron signal. The scintillator size, ND filtering, x-ray attenuation, and bias voltage were optimized for the CESZAR device so that the voltage output by the PMT was the right order of magnitude to be recorded on the oscilloscope and not saturate the PMTs. Time traces digitized by the oscilloscope were analyzed to calculate the neutron energy of the source, as shown in Figure 15. Sets of gain matched detectors were placed at the same distance from the source but at different angles in an effort to observe anisotropy in the neutron spectrum. The neutron signal is incident on the detectors at the same time, indicating the same neutron energy, but with different intensities. This may be caused by the source, or by scattering in attenuating material between the source and detector, such as the flanges on the vacuum chamber.

To aid in the selection of an appropriate location for the scintillator detectors, MCNP6 simulations of neutron scattering were conducted. To quantify the amount of scattering the h -factor, i.e. the number of neutrons including scattering divided by number of neutrons without scattering, was calculated [33]. Figure 16 shows a pseudo-color h -factor plot, which illustrates a “lighthouse” like feature that is created from neutron scattering within the electrodes. An h -factor greater than one indicates neutrons scattered into the area, while an h -factor less than one indicates neutrons were scattered away and there is a lower incident neutron flux in the spatial region.

The time of the pinch was determined from the x-ray arrival time in all the detectors, subtracting the photon time of flight. Signal recorded at approximately the same time in detectors at different distances was used to determine the time of x-ray emission, indicated by the solid black line in Figure 15. Assuming that the neutrons are born at the same time as

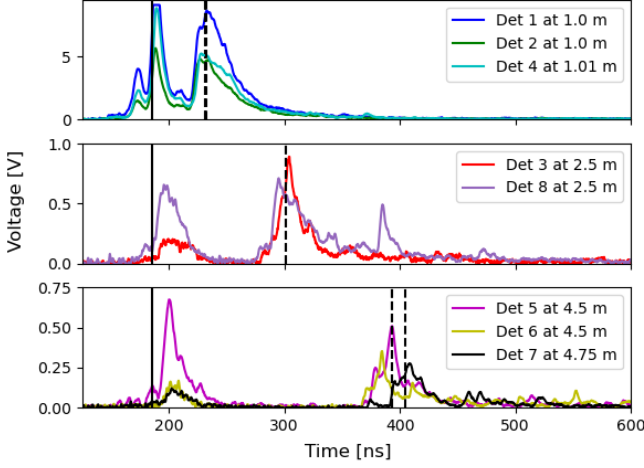


Fig. 15. Neutron TOF traces of the same CESZAR shot for detectors located at (top) 1 m, (middle) 2.5 m, (bottom) 4.5 and 7 m, as indicated in the legend. X-ray birth time (solid black) and arrival time of 2.45 MeV neutrons (dashed black) are shown as vertical lines.

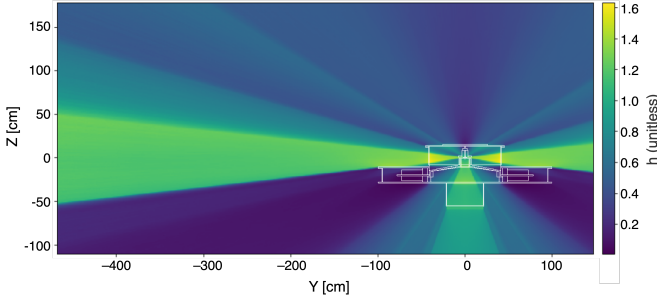


Fig. 16. Pseudo-color plot of the scattering h -factor at the CESZAR device. The Z-axis corresponds to up/down in the physical world. Neutron emission is centered at (0,0) and the outline of the device geometry is shown with white lines.

the hard x-rays, the predicted arrival time of a 2.45 MeV neutron is calculated for detectors at different distances from the source, indicated in the dashed black lines in Figure 15. The measured signals are consistent with the neutron energy from D-D fusion.

IV. CONCLUSIONS AND FUTURE WORK

A Portable and Adaptable Neutron Diagnostics for Advancing Fusion Energy Science (PANDA-FES) was designed, assembled, and shipped to different facilities. The total configuration of the system consists of three sets of 8 plastic scintillator detectors (24 total detectors) and 3 LaBr activation detectors. The plastic scintillator sets can be operated together or separately, as can the LaBr activation detectors. These systems were used to measure the neutron yield and time dependent neutron signals from CESZAR and FuZE. Experimental campaigns were run to measure the neutron energy-angle distribution on CESZAR and the deuteron beam energy and spatiotemporal source on FuZE. This diagnostic system has been successful on experiments with time scales from 10s of ns to 10s of μ s, with neutron yields on the order of 10^6 to 10^{11} . The portable system can be tailored to various types of

D-D fusion experiments to characterize different parameters relevant to plasma physics.

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